

FUNCTIONAL RANDOM FOREST FOR MIXED DATA

Mohammad Fayaz

Alireza Abadi

Shahid Beheshti University of Medical Sciences

Tehran, Iran

Review

Methods

Computations

Results

Conclusion

Outline

Review

Methods

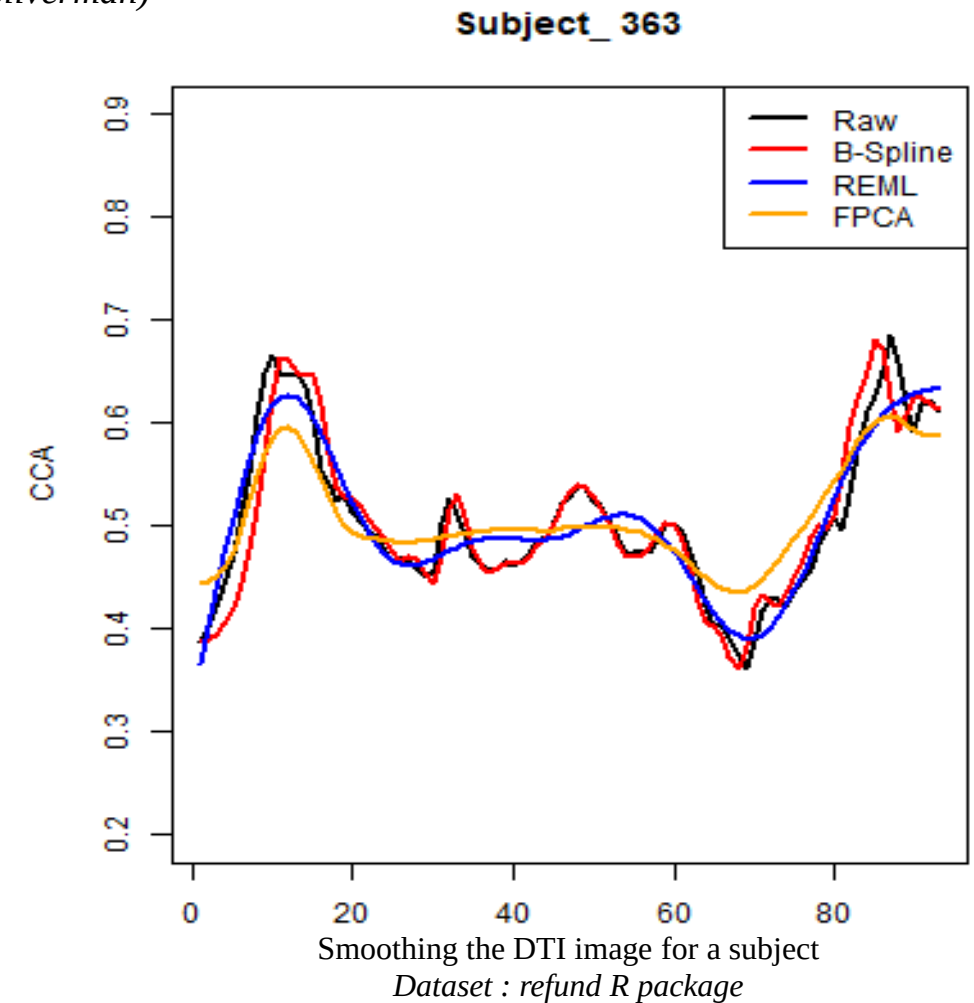
Computations

Results

Conclusion

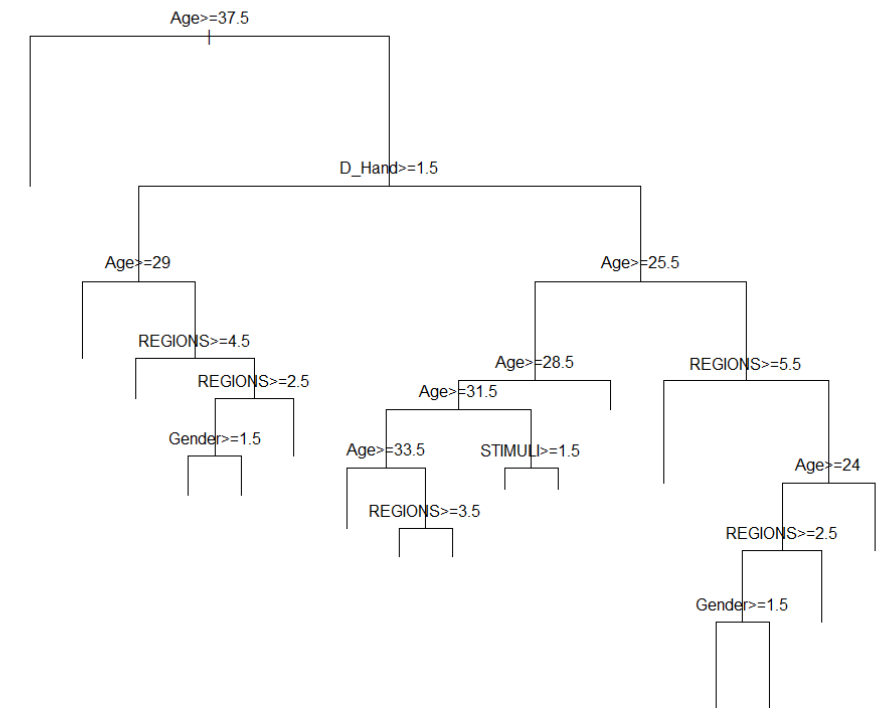
Functional Data Analysis

- Branch of statistics dealing with high dimension data (*Ramsay and Silverman*)
 - Time series
 - Images
 - And etc.
- Representing data as smooth curves
$$\{x_n(t): t \in [T_1, T_2], n = 1, 2, \dots, N\}$$
- Dimension reduction methods:
 - B-Spline basis functions
 - Fourier Series basis functions
 - Functional Principal Component Analysis
 - And etc.
- The regression models :
 - Scalar-on-Function Regression (SOF)
 - Function-on-Scalar Regression (FOS)
 - Function-on-Function Regression (FOF)



Statistical Learning

- The science of learning from data in: (*Hastie, Tibshirani and Friedman*)
 - Statistics
 - Data mining
 - Artificial Intelligence
- Models:
 - Supervised
 - Prediction
 - Classification
 - Unsupervised
- Linear Discriminate Analysis (LDA)
- Support Vector Machine (SVM)
- Classification and Regression Tree (CART)
- **Random Forest (RF)**



CART model outputs
with Region of Interest (ROI)
in the Brain

Random Forest in Functional Data

- **Question ?**
 - **Functional data in the Random Forest ? How?**
- **Scalar-on-Function Regression:**
 - Random Splitting (*Möller et. al.*) (*Pospisil & Lee*)
 - Basis function coefficients (*Ramsay & Silverman*) (*Gregorutti et. al*)
- **Function-on-Scalar Regression:**
 - Basis function coefficients (*Yu & Lambert*)
 - Density Functions ,
 - using Kullback-Leibler divergence,
 - Hellinger and Euclidean distance (*Nerrini & Ghattas*)
- **Function-on-Function Regression:**
 - Generalized B-Spline,
 - using Kullback-Leibler divergence,
 - Hellinger and Euclidean distance (*Rahman et al.*)

- **Question ?**
 - **What mixed data means in FDA?**
- **Mixed or Hybrid Data**

*“It is a characteristic of statistical methodology that problems do not always fall into neat categories. ... We often have **both a vector** of data and **an observed function** on each individual of interest.*

Suppose that our observations consist of pairs (x_i, y_i)

- *x_i is a **function** on the interval T*
- *y_i is a **vector** of length M .*
- *For example: Canadian Weather Data Stations” (Ramsay & Silverman)*

Functional Regression with Mixed Data

- Question ?
 - How to incorporate mixed data in the functional regressions?

- **Semi-functional partially linear model** (*Aneiros-Pe' rez & Vieu*)

$$Y = r(X_1, \dots, X_p, T) + \varepsilon = \sum_{j=1}^p X_j \beta_j + m(T) + \varepsilon,$$

- Real explanatory variables: $X_j (j = 1, \dots, p)$
- Functional nature of explanatory variable: T

- **Partial Functional Linear Regression** (*Shin*)

- Let Y a r.v.
- Z is p-dimensional vector of r.v.
- $X(t)$ be stochastic process

Like principal components regression (PCR)

- And ...

$$Y_i = \beta^T z_i + \int_0^1 \gamma(t) X_i(t) dt + \varepsilon_i, \quad i = 1, \dots, n$$

Karhunen-Loève representation

$$Y_i = \beta^T z_i + \sum_{j=1}^m \gamma_j \langle X_i, \phi_j \rangle + \varepsilon_i \quad i = 1, \dots, n$$

$$\gamma_j = \langle \gamma, \phi_j \rangle$$

How model
mixed data in
functional
random forest ?

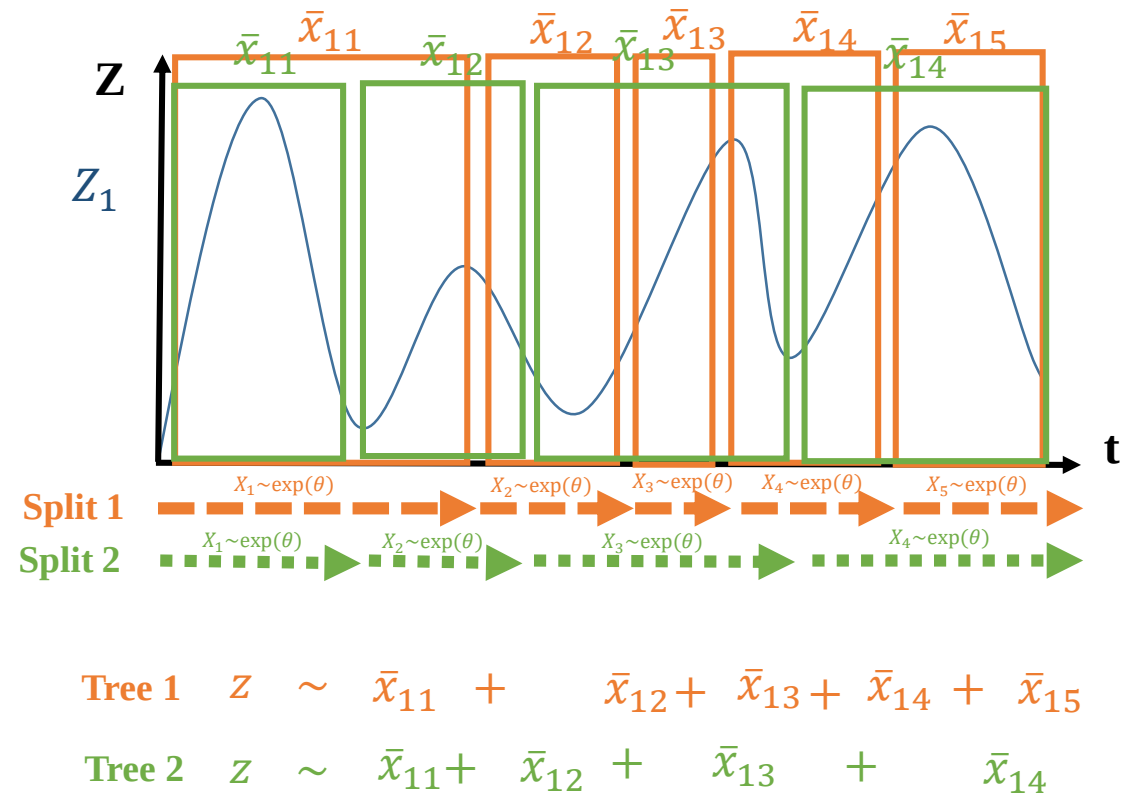
Random Splitting Approach

- **Scalar-on-Function Regression:**
 - Random Splitting (Möller et. al.) (Pospisil & Lee)
 - Common approach in machine learning models

$$Z_i(t_j), \quad j \in \{1, \dots, M\} \quad i \in \{1, \dots, N\}$$

$$X_1 \sim \exp(\theta) \xrightarrow{[t_1, t_1 + X_1]} \bar{x}_{i1} = \sum_{i=1}^N \sum_{t=t_1}^{t_1+X_1} \frac{Z_i(t)}{(t_1 + X_1 - t_1)}$$

$$X_2 \sim \exp(\theta) \xrightarrow{(t_1 + X_1, t_1 + X_1 + X_2]} \bar{x}_{i2} = \sum_{i=1}^N \sum_{t=t_1+X_1}^{t_1+X_1+X_2} \frac{Z_i(t)}{(t_1 + X_1 + X_2 - t_1 + X_1)}$$



Functional Principal Component Analysis

The principal component scores

$$f_i = \int \beta x_i = \int \beta(s)x_i(s)ds.$$

$\beta(s)$: Weight functions

$x_i(s)$: observed functions

First Step

1. Find the weight vector $\xi_1 = (\xi_{11}, \dots, \xi_{p1})'$ for which the linear combination values

$$f_i = \frac{1}{N} \sum_i \left(\int \xi_1 x_i \right)^2 = \xi_1' x_i$$

have the largest possible mean square $\frac{1}{N} \sum_i f_{i1}^2$ subject to the constraint

$$\int \xi_1(s)^2 ds = 1$$

Second Step

2. Doing the same for subsequent steps (max number of variables)

For m th step

Compute a new weight vector ξ_m

With components ξ_{jm} and new values $f_{im} = \xi_m' x_i$

f_{im} have maximum mean square, subject to the constraint $\|\xi_m\|^2 = 1$ and $m - 1$ additional constraint(s)

$$\sum_j \xi_{jk} \xi_{jm} = \xi_k' \xi_m = 0, \quad K < m.$$

(Ramsay & Silverman)

Hybrid Principal Component Analysis

The Hybrid Data

Suppose that our observations consist of pairs (x_i, y_i)

- x_i is a function on the interval T
- y_i is a vector of length M .
- $z = (x, y)$
 - x is a smooth function.
 - y is a vector of length M .
- The inner-product of
 - $z_{(1)} = (x_{(1)}, y_{(1)})$ &
 - $z_{(2)} = (x_{(2)}, y_{(2)})$ is
 - $\int x_{(1)} x_{(2)} + y'_{(1)} y_{(2)}$.
- Norm $\|z\|^2 = \langle z, z \rangle$

The Hybrid PCA

Weight functions (ξ, v)

The PC score $\eta_i = \int x_i(s) \xi(s) ds + y'_i v$.

First Step

Find $\zeta = (\xi, v)$ in Z to maximize the sample variance of the $\langle \zeta, z_i \rangle$

Subject to $\|\zeta\|^2 = 1$.

Second Step

The next pc maximize the same sample variance subject to the additional condition of orthogonality of the PCs.

SOF Random Forest with Random Splitting in the Mixed Data

Model Specification

We develop

- scalar-on-function random forest
- with random splitting (*Möller et. al.*)
- mixed data (functional and scalar covariates)

The variables:

z is a vector,

$x(t)$ is a observed function.

y is a vector of length M .

The relationship: $z = x(t) + y$.

Example:

- z is a dichotomous response
(Case / Control)
- $x(t)$ is a observed function covariate.
(fractional anisotropy tract profiles from the corpus callosum, cca(s))
- y is an integer covariate
(The number of visits) .

Algorithm SOF Random Forest with Random Splitting for Mixed Data

1. For $b = 1$ to B
 1. Split the curve $x(t)$ randomly with one of the following distributions: Exponential, Normal and Uniform.
 2. Calculate one of the statistics for each random splits: Mean, Median, Range, Min, Max, Quantile 25 and Quantile
 3. Convert $x(t)$ to the $\bar{x}_1(t), \dots, \bar{x}_p(t)$
 4. Draw a bootstrap sample h^* of size N from the training data.
 5. Grow a random-forest tree T_b with the bootstrap data, by recursively repeating the following the steps:
 1. Import the variable from $\bar{x}_1(t), \dots, \bar{x}_p(t)$ and y .
 2. Pick the best variable/split-point among them.
 3. Split the node into two daughter nodes.
2. Output the ensemble trees $\{T_b\}_1^B$.

To make the prediction at a new point:

Regression: $\hat{f}_{rf}^B(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$

Classification : Let $\hat{C}_b(x)$ be the class prediction of the b th random-forest tree.

Then $\hat{C}_{rf}^B(x) = \text{majority vote}\{\hat{C}_b(x)\}_1^B$.

The red lines are different part from random forest algorithm. (Hastie, Tibshirani & Friedman) 11

SOF Random Forest with Random Splitting in the Multiple Mixed Data

Model Specification

We develop

- The scalar-on-function random forest
- with random splitting (*Möller et. al.*)
- **Multiple** mixed data (functional and scalar covariates)

The variables:

z is a vector,

$x_1(t), \dots, x_k(t)$ are observed function.

y is a vector of length M .

The relationship: $z = x_1(t) + \dots + x_p(t) + y$.

Example:

- z is a dichotomous response
(STIMULI Standard vs Target)
- $x_1(t), \dots, x_p(t)$ is are observed function covariate.
(Region of interests (ROI)s in the brain)
- y is an integer covariate
(demographic variables such as Age, Gender, Dominant Hand and ...).

Algorithm SOF Random Forest with Random Splitting for Mixed Data

1. For $b = 1$ to B
 1. For $k = 1$ to K
 1. Split the curve $x_k(t)$ randomly with one of the following distributions: Exponential, Normal and Uniform.
 2. Calculate one of the statistics for each random splits: Mean, Median, Range, Min, Max, Quantile 25 and Quantile
 3. Convert $x_k(t)$ to the $\bar{x}_{k1}(t), \dots, \bar{x}_{kp}(t)$
 2. Draw a bootstrap sample h^* of size N from the training data.
 3. Grow a random-forest tree T_b with the bootstrap data, by recursively repeating the following steps:
 1. Import the variable $\bar{x}_{11}(t), \dots, \bar{x}_{1p}(t), \dots, \bar{x}_{k1}(t), \dots, \bar{x}_{kp}(t)$ and y .
 2. Pick the best variable/split-point among them.
 3. Split the node into two daughter nodes.
2. Output the ensemble trees $\{T_b\}_1^B$.

To make the prediction at a new point:

Regression: $\hat{f}_{rf}^B(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$

Classification : Let $\hat{C}_b(x)$ be the class prediction of the b th random-forest tree.

Then $\hat{C}_{rf}^B(x) = \text{majority vote}\{\hat{C}_b(x)\}_1^B$.

The red lines are different part from random forest algorithm. (Hastie, Tibshirani & Friedman) 12

Computations

Simulation SOF Random Forest with Random Splitting for Multiple Mixed Data

Simulated Dataset

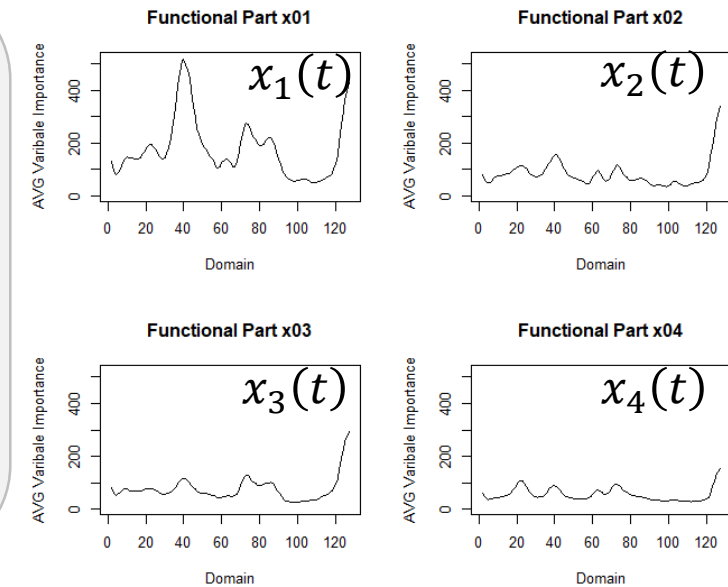
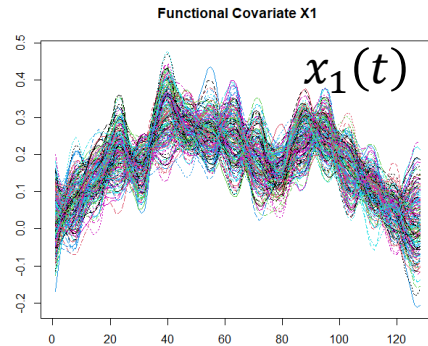
Y : Response has normal distribution
 $x_1(t), \dots, x_{20}(t)$: Functional Covariates

It showed that the only first 8 functional covariates are predictive [Gregorutti].

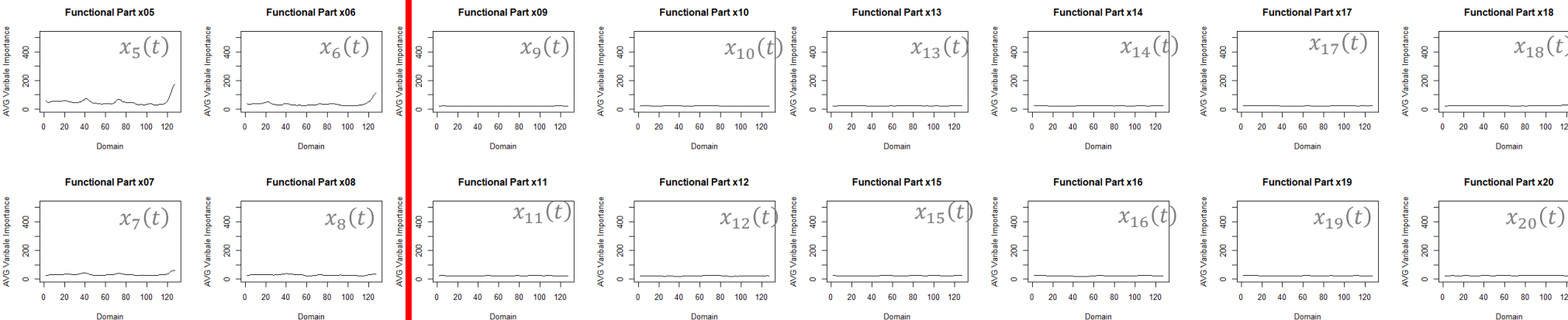
We not only show which functional covariates are predictive, we also show which part of the each functional covariate has higher variable importance than other parts.

Result

Statistics	In Bag	OOB
Correlation	0.95	0.78
Mean Square Error	0.687	3.290



These Functional Covariates are not predictive



Conclusion

The variable importance plot shows that the first 8 curves have values more than 0.

It also shows that which part of each curve has higher value of variable importance.

FOS Random Forest with Hybrid PCA

Model Specification

We develop

- function-on-scalar random forest
(Nerrini & Ghattas) (Rahman et al.)
- mixed data (functional and scalar covariates)

The variables:

$z(t)$ is an observed function,
 $\mathbf{x}$ is a vector of length M.
 $\mathbf{y}$ is a vector of length K.

The relationship: $z(t) + \mathbf{x} \sim \mathbf{y}$.

The Hybrid PCA between $z(t)$ and $\mathbf{x} \sim \mathbf{y}$

Example:

- z is an observed function
(ERP Waves – Evoked Related Potentials)
- $\mathbf{x}$ is a vector.
(Region of the brain)
- $\mathbf{y}$ is a covariate
(Stimuli , dominant hand) .

Algorithm *FOS Random Forest with Hybrid PCA*

1. For $n = 1$ to N_{Total}
 1. Calculate and estimate the hybrid PCA for response curve and vectors by (Scheffler et al.) and (Ramsay & Silverman).
2. Converting data from $(z(t), \mathbf{x}, \mathbf{y})$ to $(\eta, \mathbf{y})$.
3. For $b = 1$ to B
 1. Draw a bootstrap sample K^* of size N from training data.
 2. Grow a random-forest for **function-on-scalar tree (Nerrini & Ghattas)** T_b to the bootstrapped data, by recursively repeating the following steps for each terminal nodes of the tree, until the minimum node size n_{min} is reached.
 1. Select m variables at random from the p variables.
 2. Pick the best variable/ split-point among the m .
 3. Split the node into two daughter nodes.
4. Output the ensemble trees $\{T_b\}_1^B$.

To make the prediction at a new point:

Regression: $\hat{f}_{rf}^B(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x})$

Classification : Let $\hat{C}_b(\mathbf{x})$ be the class prediction of the b th random-forest tree. Then $\hat{C}_{rf}^B(\mathbf{x}) = \text{majority vote}\{\hat{C}_b(\mathbf{x})\}_1^B$.

The red lines are different part from random forest algorithm. (Hastie, Tibshirani & Friedman) 14

FOF Random Forest with Splitting and distance based criteria for Mixed Data

Model Specification

We develop

- function-on-function random forest. (Möller et al.)
- Random splitting for functional covariates
- FOS tree for functional response (Rahman et al.) (Nerrini & Ghattas)
- Hybrid PCA for functional and vector part

The variables (*Example: EEG-fMRI*):

- *Responses:*
 - $z(t)$ is an observed function, (ERP)
 - x is a vector of length M. (Region of head)
- *Covariates:*
 - $u(t)$ is an observed function (ROIs)
 - y is a vector of length K. (Demographics)

The relationship: $z(t) + x \sim u(t) + y$.

The Hybrid PCA between $z(t)$ and $x \sim u(t) + y$

Algorithm FOF Random Forest with Hybrid PCA

1. For $n=1$ to N_{Total}
 1. Calculate and estimate the hybrid PCA for response curve and vectors by (Scheffler et al.) and (Ramsay & Silverman).
2. Converting data from $(z(t), x, u(t), y)$ to $(\eta, u(t), y)$.
3. For $b=1$ to B (Number of trees)
 1. For $k=1$ to K (Number of predictive curves)
 1. Split the curve $u_k(t)$ randomly with one of the following distributions:
Exponential, Normal and Uniform.
 2. Calculate one of the statistics for each random splits:
Mean, Median, Range, Min, Max, Quantile 25 and Quantile
 3. Convert $u_k(t)$ to the $\bar{u}_{k1}(t), \dots, \bar{u}_{kp}(t)$
 2. Draw a bootstrap sample h^* of size N from the training data.
 3. Grow a random-forest tree T_b with the bootstrap data, by recursively repeating the following the steps:
 1. Import the variable $\bar{u}_{11}(t), \dots, \bar{u}_{1p}(t), \dots, \bar{u}_{k1}(t), \dots, \bar{u}_{kp}(t)$ and y .
 2. Pick the best variable/split-point among them.
 3. Split the node into two daughter nodes.
4. Output the ensemble trees $\{T_b\}_1^B$.

To make the prediction at a new point:

Regression: $\hat{f}_{rf}^B(x) = \frac{1}{B} \sum_{b=1}^B T_b(x)$

Classification: Let $\hat{C}_b(x)$ be the class prediction of the b th random-forest tree.
Then $\hat{C}_{rf}^B(x) = \text{majority vote}\{\hat{C}_b(x)\}_1^B$.

The red lines are different part from random forest algorithm. (Hastie, Tibshirani & Friedman) 15

Simulation - Comparing unsmooth vs smooth curves in the prediction accuracy

Method

Goal

Predict the visitor status based on the brain image

Response (z)

Visitor status (Health/ multiple sclerosis)

Functional Covariate ($x(t)$)

The Fractional anisotropy tract profiles from the corpus callosum (cca)

Model

$$z \sim x(t)$$

Method

SOF random forest with random splitting.

Data Reference

DTI dataset in the refund R package.

Setting

- Number of trees: 250, 500 , 750 and 1000
- Random splitting values generate from exponential distribution with the following parameters: 0.1, 1 and 10
- Compared
 - Overall Accuracy,
 - Sensitivity and
 - Specificity

Result

Index	Unsmooth	B-Spline (GCV)	Spline (REML)	FPCA
Sensitivity	0.238	0.143	0.19	0.976
Specificity	0.982	0.976	0.857	1
Overall Accuracy	0.899	0.883	0.888	0.984

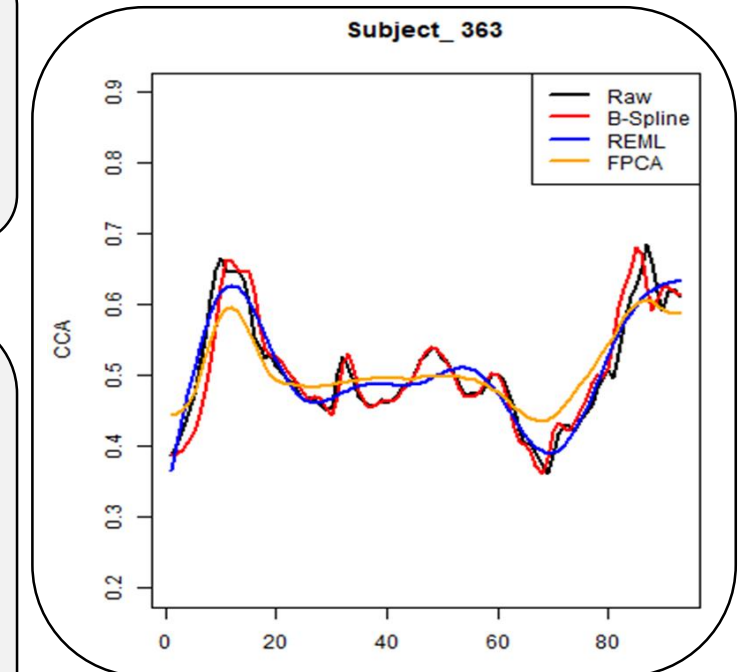
Best setting: number of tree 250 and exponential parameter 1.

Comparing

1. Unsmooth
2. B-Spline (GCV)
3. Spline (REML with GAM)
4. FPCA (First fPC)

Conclusion

- 1- Random splitting has not high sensitivity with unsmooth data.
- 2- Smoothing the curves dose not improve the model.
- 3- FPCA, as smoothing method, improve the sensitivity of the random splitting approach.



All smooth curves are calculated for each visitor and shown in the R-Shiny App

Comparing existing methods for functional statistical learning models with and without mixed data

Method

Goal

Predict the visitor status based on the brain image

Response (z)

Visitor status (Health/ multiple sclerosis)

Functional Covariate ($x(t)$)

The Fractional anisotropy tract profiles from the corpus callosum (cca)

Scalar Covariate (y)

The number of visits.

Models

1) $z \sim x(t)$

2) $z \sim x(t) + y$

Method

Functional statistical learning methods

The R Package

The fda.usc (Manuel Febrero Bande & Manuel Oviedo de la Fuente)

Data Reference

DTI dataset in the refund R package.

Comparing

CART: Classification and Regression Tree

RF: Random Forest

NNET : Neural Network

SVM: Support Vector Machine

LDA : Linear Discriminate Analysis

NB: Naive Bayes

Conclusion

1- RF models has low sensitivity.

2- RF models with mixed data improve the sensitivity of the models.

3- Naïve Bayes models have the best performance in two scenrios.

Model 1: One Functional Covariate

Train	Methods					
	CART	RF	NNET	SVM	LDA	NB
Sensitivity	0.769	0.269	0.962	0	0.885	0.769
Specificity	0.949	0.98	0.984	1	0.812	0.809
Overall	0.933	0.915	0.982	0.908	0.819	0.805

Test	Methods					
	CART	RF	NNET	SVM	LDA	NB
Sensitivity	0.312	0.125	0.312	0.062	0.562	0.438
Specificity	0.91	0.962	0.936	0.987	0.782	0.744
Overall	0.809	0.819	0.83	0.83	0.745	0.691

Model 2: One Functional and Scalar Covariate

Train	Methods					
	CART	RF	NNET	SVM	LDA	NB
Sensitivity	0.769	0.423	0.846	0.077	0.923	0.962
Specificity	0.965	0.988	1.000	1.000	0.805	0.816
Overall	0.947	0.936	0.986	0.915	0.816	0.830

Test	Methods					
	CART	RF	NNET	SVM	LDA	NB
Sensitivity	0.313	0.125	0.375	0.188	0.750	0.938
Specificity	0.936	0.974	0.962	0.974	0.795	0.808
Overall	0.830	0.830	0.862	0.840	0.787	0.830

SOF Random Forest with random splitting with Mixed Data

Method

Goal

Predict the visitor status based on the brain image

Response (z)

Visitor status (Health/ multiple sclerosis)

Functional Covariate ($x(t)$)

The Fractional anisotropy tract profiles from the corpus callosum (cca)

Scalar Covariate (y)

The number of visits.

Model

$$z \sim x(t) + y$$

Method

SOF random forest with random splitting with Mixed Data.

Data Reference

DTI dataset in the refund R package.

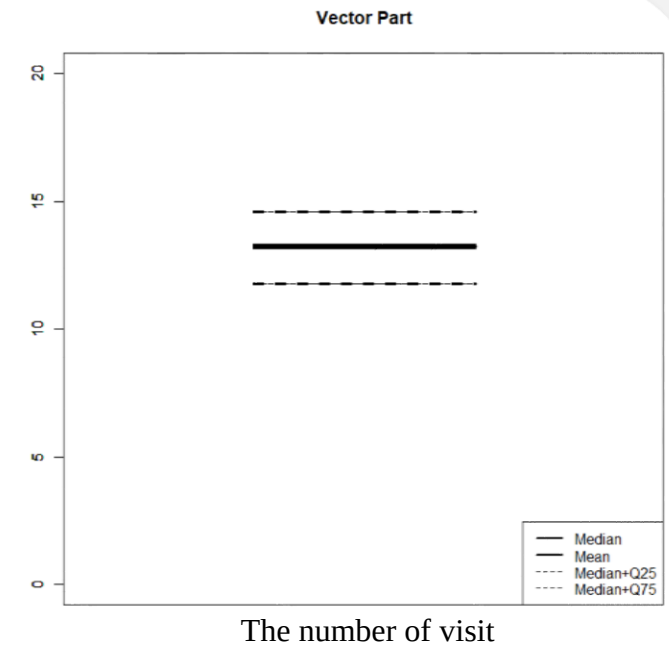
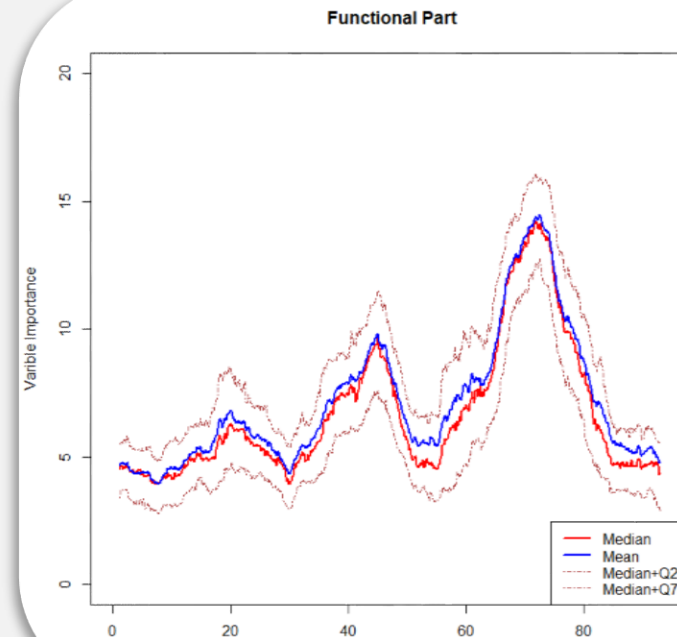
Result

Train	Splitting	
	No Mixed	One mixed
Sen	0.238	0.4
Spe	0.982	0.97
Overall	0.899	0.907

Conclusions

- 1- Adding scalar covariate improve the model performance.
- 2- The variable importance plot for functional and vector part are presented below.

The Variable Importance Curve for mixed covariates



This plot is presented in the shiny app with details

FOS Random Forest with Hybrid PCA

Method

Goal

Which covariates are related to the ERP?

Response ($z(t)$)

ERP (Evoked Related Potentials) wave in time domain [0,800] for two Stimuli of Auditory Task (Standard and Target)

Scalar Covariates (y)

The Age, Dominant Hand [R-L], Gender[M-F], Stimuli.[S-T]

The Region of Brain covariate

The region of brain in model (1) enter as covariate and in model (2) enters as vector part of functional response.

Model

$$1) z(t) \sim y + Region$$

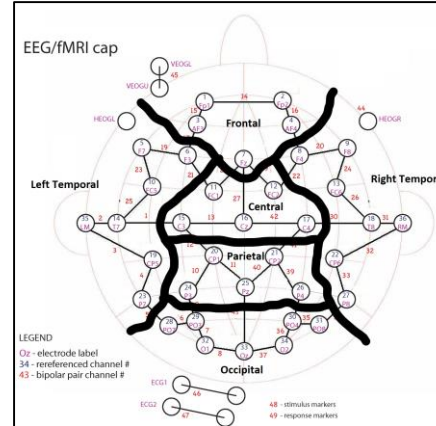
$$2) z(t) + Region \sim y \Rightarrow \eta \sim y$$

Method

FOS Random Forest with HPCA

Data Reference

EEG/fMRI Dataset from (Walz et al.)



From Walz et al.
The EEG Cap regions.

Results

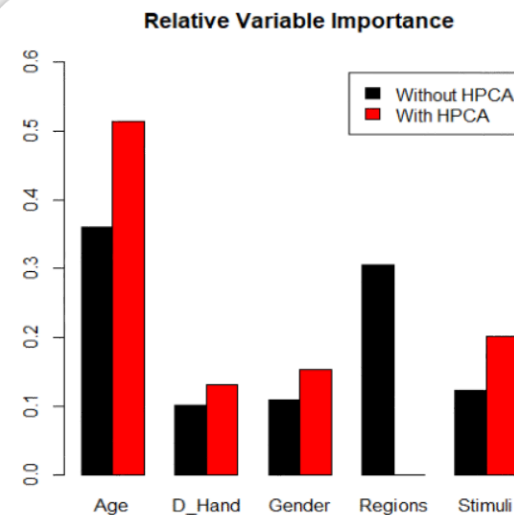


figure 1

Average based on 250,500,750 & 1000 trees

Conclusions

- 1- The region of the brain is an important covariate in studying the ERP of the whole brain.
- 2- To understand the effect of region on ERP, the first model shows high variable importance as covariate. (figure 1)
- 3- In the second model, the region with ERPs are analyzed together with HPCA as enter the model as response. The prediction error is statistically lower than the first model. (figure 2)

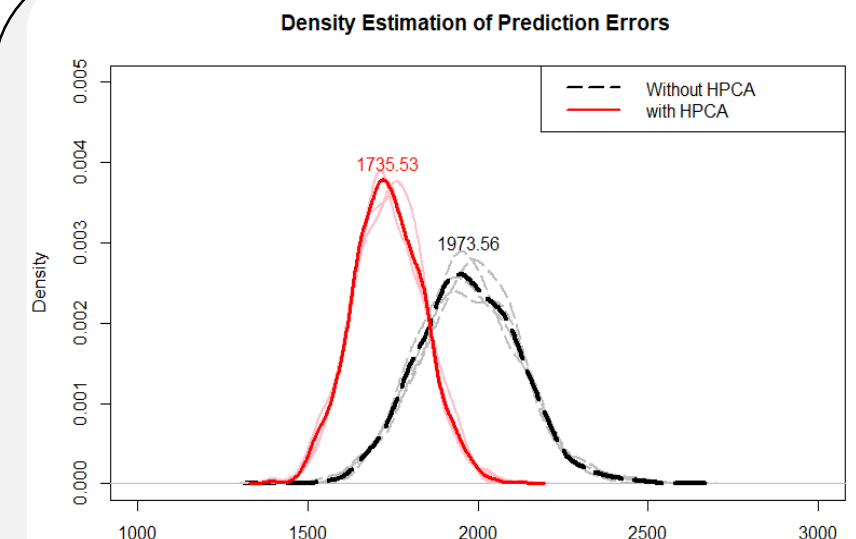


figure 2

Average based on 250,500,750 & 1000 trees

FOF Random Forest with Hybrid PCA

Method Goal

The relationship between EEG-fMRI

Response ($z(t)$)

ERP (Evoked Related Potentials) wave in time domain [0,800]ms for two Stimuli of Auditory Task (Standard and Target)

Scalar Covariates (y)

The Age, Dominant Hand [R-L],

Gender[M-F] , Stimuli.[S-T], Region [like previous model]

Functional Covariates ($x(t)$)

More than 40 Region of Interest (ROIs) time series extracted from fMRI dataset.

Model

$$1) \quad z(t) \sim x(t) + y + Region$$

$$2) \quad z(t) + Region \sim x(t) + y \Rightarrow$$

$$HPCA \quad \eta \sim x(t) + y$$

Method

FOF Random Forest with HPCA

Data Reference

EEG/fMRI Dataset from (Walz et al.)

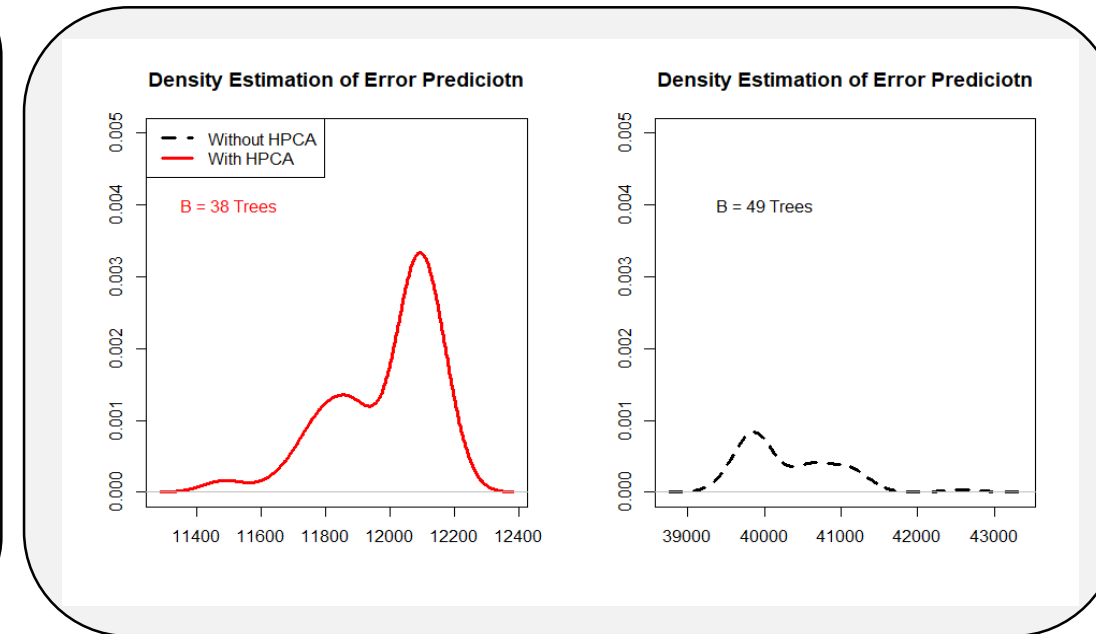
Result

The prediction errors of two methods without HPCA and HPCA compared with T-Test and Wilcoxon test. They are significant at 0.05. The HPCA method has lower median and mean than without HPCA method.

Test Name	P-Value
Welch Two Sample t-test	< 0.00
Wilcoxon rank sum test	< 0.00

Conclusion

- 1- The FOF Random Forest works well under $P > N$ conditions. P is the number of functional and scalar covariates , here is 41 functional ROIs and 4 scalar demographic variables. And N , number of subject is less than 35.
- 2- The HPCA reduces the prediction errors in model (2) compared to the model (1).
- 3- The main limitation of this method is that the variable importance curve plots can not estimate for whole time ranges with low number of trees and need at least 250 trees.
- 4- We can also analysis the response functions with random splitting [Möller] or FPCA [Yu] basis and use multivariate random forest instead of [Nerini] FOS RF. [under development]



Conclusion

Splitting & FPCA

- 1- Combining functional data analysis methods with statistical learning models makes new models for studying complex relationship and high-dimensional dataset .
- 2- Splitting is a common strategy in machine learning models while functional PCA is a core method for FDA in dimension reduction.
- 3- Splitting is easy-to-implement method but the results are correlated.
- 4- FPCA is statistical method with underlying assumptions and need statistical inference for different type of data, but the results are orthonormal.

Random Forest & Functional Regression

- 1- The random forest model the nonlinear relationships and interaction effects.
- 2- The random forest main problems are overfitting and bias output. But the results are easy to understand with variable importance plot. The functional variable importance plot show which part of the curves are more important than others.
- 3- The functional random forest provides a way for analyzing functional data with random forest algorithm.

Computations & Algorithms

- 1- The functional random forest algorithms are not well studied like functional data analysis methods and fast computations with efficient algorithms are in attention.
- 2- The variation of functional and scalar part in the hybrid data summarizes with hybrid PCA which reduce the prediction errors.
- 3- The functional random forest with mixed data are developed for multiple functions and scalars, with random splitting approach and with hybrid PCA.

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Departments	• Department of Biostatistics, Shahid Beheshti University of Medical Sciences, Tehran, Iran. • Department of Biostatistics, Tehran University of Medical Sciences, Tehran, Iran. • Department of Statistics and Biostatistics, Tarbiat Modares University , Tehran, Iran. • National Brain Mapping Lab, Tehran, Iran. • Institute for Research in Fundamental Sciences, Tehran, Iran.
Workshops	• Functional Data Analysis Summer School, Compstat 2018, Iasi, Romania. • eRum 2020 , virtual event, Milan, Italy. • EEG Data Analysis, NBML, 2019-2020, Tehran, Iran. • The overview of Dimension reduction and Functional Data Analysis, 2017, Tarbiat Modares University , Tehran, Iran.
People	• David Nerini • Iman Mohammad-Rezazadeh 